

Distributed computing GIK2NX



Version 3.0

Innehåll

Prerequisite	2
Purpose.....	2
Deadline	2
How to study	2
Aid	2
Tools	2
Submitting the lab	2
Grades	3
Lab 1	3
Documentation and research	3
Summary of documentation:.....	3
Coding - Create an FSM game!	4
Recording.....	4

Prerequisite

You need to have SPADE up and running before the lab and checked that the Hello Dummy Agent program works.

Purpose

This lab is mostly about understanding the foundations of the framework of SPADE and the basic behaviors in an agent environment. But also finding peer reviewed references, debating its validation and use its content for explaining/deepening knowledge about the topic.

Deadline

See Canvas.

How to study

This lab is solved within the groups.

Aid

Everything, but watch out for plagiarism. Plagiarism is the shortness to hand in on someone else's work as their own. Obviously, if it is a groups work, then you will have the same solution, but all your names need to be on the solution. In many cases, plagiarism, this can happen unconsciously, and that is why you have to be careful and conscious about how to use external sources.

Tools

IDEA for coding with Python as programming language and SPADEs framework.

Submitting the lab

When submitting, apply all options:

You choose whether you want to write text/Code in Swedish or English

All articles that are referenced must be fully downloaded, saved and enclosed in your submission. (This is to simplify and speed up the correction procedure)

Save your answers + referenced articles *as PDFs* and complete working code compressed in one zip.

Also **a recording** – provide the recording uploaded on “Play(My Media)” – see the guide for Play(My Media). (see further down in the document also) Another option is to use YouTube and hand in the link for the recording in the report or directly in text-editor ! (The recording cannot be submitted as a regular file in the submission due to lack of space in the course room.)

Name the zip with your *group number followed by lab number*, for example, group1lab1.zip in Canvas. Your submission time is automatically registered when you submit.

Grades

U – Fail

G- Pass, all requirements must be met and correct

Lab 1

Documentation and research

It should be a minimum of 2 pages and maximum 4 pages long excluding references with a font size of 11 or 12. Use APA standard as reference, there is a reference tool in word you can use if you want to.

On the first page make sure all the group members names are written!

Find one (or more) peer reviewed article or report that can be associated with this lab (possible keywords: SPADE, Agents, Software Agents, Agent behavior, FSM, Agents states).

Make a short explanation of the article/reports content and conclusion. Make sure it's relevant to the topic of software agents. An unrelated article/report will not be accepted.

Make a connection to the lab or your reflections about the topic, use course literature, lectures, lessons or exercises to what you just read. If there is no link between what you have read and what you have learned write about that, how is it different?

Argue about the article/report's validation, is it reliable or bias? Is it trustworthy?

Explain the coding game, how did you use the FSM and different states? What is the goal of the game?

Create a model for your FSM program that show the different states and their relations.

End the documentation with a reference list on a new separate page (which is excluded from the 2 to 4 pages of text)

Summary of documentation:

- Names
- Explain an article/report
- Reflect on how that article/report is relevant to your studies
- Argue about the article/report's validation
- Explain your coding approach and usages of FSM
- Explain the games nature/goal
- Include a picture/model for the FSM program
- End with reference list

Coding - Create an FSM game!

Implement an FSM with minimum of **six** states including start and end. You will need to use the advanced behaviors [from SPADE framework](#)

Create a game, use your imagination. Earlier years we've had games about a woodchopper, cutting down wood before it already was downed by birds. A knight fighting different bosses. A runner trying to catch the bus and so on. Some student added pictures based on either victory or defeat, some added randomized values, so the games never ended the same way, making it interesting every time.

The FSM Machine can easily "only be a counter", state X goes to Z and increases every time, when X reaches a limit, it goes to state Y. Done. *Well, this lab is about understanding the behaviors and applying your own touch and knowledge so make it a game not just a counter.*

The game could work independently or ask/prompt the user for inputs, it's up to you.

Comment the code on a level as you would explain it to a fellow student, I don't want to read "this is a string, string is when letters are used" or summarization such as: This is the state A. This is state B end. Make it the way you would have wanted it to be if you were to take over it from another group.

Make sure all group members names are included as comments in the beginning of the code. It needs to work, if not it will not be going to pass.

Recording

End the lab by making a recording (regarding submission, see above) where all group members participate and start by introducing themselves ! When recording, explain the code (keep comments in the code but try not to just read them as text, speak freely) and make a demonstration of the code, run the game. If you use random functions, run the game at least two times so the planned effect shows.

Good luck!